Prefix Memory and the Growth Spectrum of Event Observers

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This paper is Paper XLIV (version 1.0) of the RIME program. It classifies prefix-memory event observers on the fixed circular action and determines their exact finite quotients and asymptotic growth spectrum.

Abstract

We study how retaining an initial segment of an event history changes the minimal predictive state of a circular coalescence system. The raw actions are rotation and one adjacent collapse, started from placed singletons, with registration of a fixed source pair. A depth-j observer retains the registration origin and the first j complete selected absorption records, and observes every subsequent selected event. Using the actual chain and placement-fibre classification, we give the exact minimal quotient for every positive depth. The finite classification and exact quotient cardinalities are formalized in Lean 4; the asymptotic results are proved separately. For fixed j, its state count is asymptotic to $5\,2^j\sqrt{n}\,4^n/(16\sqrt{\pi})$. For $j = \lfloor \theta n \rfloor$, the exponential rate increases strictly from 4 to 9/2 on $0 < \theta < 1/3$ and equals 9/2 for $1/3 \le \theta < 1$. A uniform comparison with a truncated binomial sum proves the growth limit, including the threshold and integer rounding. Every positive sublinear depth has base 4. Above the critical density, the prefix and complete-summary state counts are asymptotically equal, with an exponentially vanishing relative deficit. At the critical density, a square-root window has a Gaussian transition profile, and the quotient ratio at depth $\lfloor n/3 \rfloor$ tends to 1/2. Thus the same actual dynamics supports strictly intermediate observer-state growth rates, with a resolved critical transition, and the full exponential rate can occur before the observer retains the full history.

Keywords: event history; prefix memory; observer-relative state minimization; circular coalescence; exponential growth spectrum

1 Introduction

How much of an old event history must a predictive observer retain? The answer depends on the report being requested. In a circular registered absorption system, the future selected-event observer can forget the old selected chain once the current placement is known, whereas a complete-summary observer reports that chain. Their exact minimal state counts have exponential bases 4 and 9/2, respectively [2]. We investigate the observations between these two contracts by retaining a prefix of the old chain.

The retained data are complete source-addressed events, not only their number or a sequence of left/right signs. A depth-j observer reads the origin, the first j selected records, and whether the retained prefix is still shorter than j. It also observes all future selected events under every common raw continuation. For fixed j, this extra memory changes a leading constant but not the growth base. At linear depth the rate changes: for $j = \lfloor \theta n \rfloor$ it is

$$\lambda(\theta) = \begin{cases} \dfrac{2^{2+\theta}}{(1+\theta)^{1+\theta}(1-\theta)^{1-\theta}}, & 0 < \theta \le 1/3, \\ 9/2, & 1/3 \le \theta < 1. \end{cases} \tag{1.1}$$

In particular the open interval $(4, 9/2)$ consists of actual growth rates of declared observers on the same dynamics. The plateau at $1/3$ is a rate statement, not equality of the underlying state quotients. Even full selected-history memory in our contract does not observe the physical phase of a rank-one packet.

For strictly supercritical depth we prove more than equality of growth bases: $Q_n^{[\lfloor \theta n \rfloor]}/Q_n^{\mathrm{full}} \to 1$. The proof counts complete summaries by selected-chain length and bounds the loss caused by long chains and forgotten terminal phases separately. At critical density the same chain-length profile gives a central-limit transition when the retained depth is shifted by order $\sqrt{n}$.

The mechanism has two parts. The exact quotient separates future behaviour from retained reports. Counting the reports requires distinguishing a saturated prefix, which may have an unobserved past continuation, from a short prefix that is the entire current chain. A uniform reduction then replaces the resulting coefficients by a truncated binomial sum up to polynomial factors. Its entropy maximum gives the rate in (1.1).

Sequential-machine equivalence supplies the general language of right indistinguishability [5, 6, 8]. We use, rather than reprove, the actual source-addressed chain and fibre classification of [3] and the constructive future separation theorem of [2, Theorem 6.2]. The new problem here is the prefix-indexed family of observation contracts, its exact count for every depth, and its uniform growth spectrum. The state numbers concern this model and these observers; the parameter j is a retained-event depth, not a raw-word budget.

2 Dynamics, Observers and Imported Structure

Fix $n \geq 3$, $Q = \mathbb{Z}/n\mathbb{Z}$, and source atoms $\Omega = \{0, \dots, n-1\}$. A placement P assigns disjoint nonempty atom packets to occupied sites, covering Ω. The singleton seed is $P_0(q) = \{q\}$. The site actions are

$$p(q) = q + 1, \qquad d(1) = 0, \quad d(q) = q \quad (q \neq 1). \tag{2.1}$$

Packets follow these maps and merge by union when their images coincide. Words are chronological. An effective fusion has complete ordered record

$$\rho = (d, 0, 1, U, V, 0, U \cup V), \tag{2.2}$$

where U, V occupied ports $0, 1$ just before the fusion.

Set $F = \{0, 1\}$. The first fusion that creates exactly F is its registration, with origin o. Once registered, a selected absorption is an effective fusion involving the current packet containing F, called the carrier. Unrelated mergers do not append a selected record. We use the full registration record (2.2) as the semantic origin o. The compact origin $(d, (0, U), (1, V), 0)$ used in the prior representation and control code is a lossless projection: its omitted result is the fixed set $F = U \cup V$. Reinserting that field recovers o. Write $x = (P, o, A)$ for the current placement, origin and selected chain. Let X_n be the set of all such states reached from the singleton seed by finite actual words. The visible output alphabet contains null, registration outputs $\mathsf{reg}(\rho)$ and selected-absorption outputs $\mathsf{abs}(\rho)$, where ρ is a complete record (2.2). On a letter a, the output $\epsilon(x, a)$ is $\mathsf{reg}(\rho)$ when that letter first creates F, $\mathsf{abs}(\rho)$ when it fuses an already registered carrier with another packet, and null otherwise. In particular, registration itself is visible; rotations, nonfusing defects and unrelated fusions emit null. Registration has result F, whereas a selected absorption has result strictly containing F, so the two event kinds are also recoverable from their full records. Write $\delta_a x$ for the actual successor. The trace $\mathrm{Trace}(x, w)$ retains one tagged output, including nulls, per letter of the common raw word w.

For $j \geq 1$ define the retained label

$$L_j(x) = \begin{cases} \bot, & x \text{ unregistered}, \\ (o, \mathrm{take}_j\, A, \mathsf{short}), & |A| < j, \\ (o, \mathrm{take}_j\, A, \mathsf{saturated}), & |A| \geq j. \end{cases} \tag{2.3}$$

The short marker reports the current chain length, not that future absorptions have ceased. The registration origin is unique for this pair when $n \geq 3$.

For completeness, define the label update U_j on reachable label/output pairs. Let $\mathsf{flag}_j(k)$ be short for $k < j$ and $\mathsf{saturated}$ for $k \geq j$. Then

$$\begin{aligned} \mathsf{U}_j(z, \mathsf{null}) &= z, \\ \mathsf{U}_j(\bot, \mathsf{reg}(\rho)) &= (\rho, (), \mathsf{short}), \\ \mathsf{U}_j((o, B, \mathsf{short}), \mathsf{abs}(\rho)) &= (o, B \parallel [\rho], \mathsf{flag}_j(|B| + 1)), \\ \mathsf{U}_j((o, B, \mathsf{saturated}), \mathsf{abs}(\rho)) &= (o, B, \mathsf{saturated}). \end{aligned}$$

Here $B \parallel [\rho]$ appends one record. Registration initializes the origin but does not enter the absorption chain; its new chain is empty and hence short because $j \geq 1$. These cases cover every actual transition, and the definitions give $L_j(\delta_a x) = \mathsf{U}_j(L_j(x), \epsilon(x, a))$.

Two states are equivalent for this observer if, under every common raw word, their selected-event traces and all retained-label reports agree, including the label at the empty prefix. Equivalently,

$$x \equiv_j y \quad \Longleftrightarrow \quad x \equiv_{\text{future}} y \ \text{ and } \ L_j(x) = L_j(y). \tag{2.4}$$

For the right-to-left implication, assume equal initial labels and equal future traces. At each letter the outputs agree, so the displayed update rule gives equal successor labels. Induction keeps the labels equal at every prefix of every common continuation. For the converse, the empty prefix gives equal current labels and agreement of all selected-event traces is exactly future equivalence. Let $Q_n^{[j]}$ count these classes on X_n, and let $Q_{n,\mathrm{reg}}^{[j]}$ count them on the successor-closed registered subset.

We record precisely the inherited structural results used below. Put $N = n - 2$.

1. **Actual chains and fibres** [3, Theorems 3.1 and 4.2]. A selected chain is uniquely encoded by nonzero checkpoints among $1, \ldots, N$ and their L/R port bindings. If its final carrier adds $r < N$ atoms to F, that carrier is one of the $r + 1$ proper arcs extending F. For a proper carrier C, list its source complement in the positive linear order starting immediately after C and ending just before it. Partition that list into nonempty consecutive intervals. The blocks C and those outside intervals must occur at their occupied physical sites in the induced positive cyclic block order. For each fixed compatible complete chain, every placement satisfying these two conditions is actually reachable. The number of these placements is $n\binom{2N-r}{N-r}$, independent of the chain. At a full carrier each complete chain has n actual physical phases. These are statements about actual words, not arbitrary record lists.

2. **Future equivalence** [2, Theorem 6.1 and Proposition 3.3]. All permanently unregistrable states and registered rank-one states are silent and form one future class. Every registerable unregistered placement is its own class. Registered nonterminal states are equivalent exactly when their current placements agree. The number of registerable placements is $R_n = n\binom{2N}{N-1}$. Constructive common-suffix probes establish the separation of these classes.

The comparator is the **complete-summary Moore observer**, whose current report is $\bot$ before registration and (P, o, A) afterwards. Write s_n for the number of actual complete registered summaries, including terminal phase. Its minimal quotient count is $Q_n^{\mathrm{full}} = 1 + R_n + s_n$

[2]: registered summaries are separated by their current report, while the unregistered region has the same dead and registerable placement classes as above.

Here a checkpoint is the cumulative number $b = |C'| - 2$ of atoms absorbed beyond F, read from that record's result C'. If the preceding carrier is at port 0, the record has binding R and absorbs a prefix of the positive ordered complement; at port 1 it has binding L and absorbs a suffix. The checkpoint increment is the size of that partner. Starting at F, these increments and bindings recover the complete ordered records. When only two packets remain the possible partner sets coincide, but the two port bindings still give different records.

Only the positive-depth contracts (2.3) are used in the main theorems. At depth zero we use the separate future-only convention, without a registration label. If that label were kept at depth zero, it would split the silent class in two, changing the whole-domain count by one.

3 Exact Prefix Classes

The extension of a proper carrier has a added atoms on its left and b on its right, with $a + b = r < N$. Complete ordered records recover these additions and the parent-to-port binding at each checkpoint. They therefore distinguish L and R even when the last two packets have identical possible partner sets.

Lemma 3.1.

Sum the number of compatible retained labels over all $r+1$ carrier arcs of extension r. The result is

$$c_{0,j} = 1, \qquad c_{r,j} = 2^j \binom{r+1}{j+1} + \sum_{\ell=1}^{j-1} 2^\ell \binom{r-1}{\ell-1} \quad (r \geq 1). \tag{3.1}$$

Binomial coefficients with an index outside their range are zero.

Proof. *First count saturated prefixes for a fixed arc (a, b). If u of the j records are left absorptions, their order has $\binom{j}{u}$ possibilities. Their cumulative endpoints on the left form a u-element subset of $\{1, \dots, a\}$; the right endpoints form a $(j-u)$-element subset of $\{1, \dots, b\}$. These data recover every ordered record. Conversely every such choice is a legal prefix, which may leave some of either side for unobserved past absorptions. The fixed-arc count is*

$$\sum_{u=0}^{j} \binom{j}{u} \binom{a}{u} \binom{b}{j-u}. \tag{3.2}$$

For each u, sum over $a + b = r$. The convolution is

$$\sum_{a=0}^{r} \binom{a}{u} \binom{r-a}{j-u} = \binom{r+1}{j+1}. \tag{3.3}$$

One proof chooses a $(j+1)$-subset of $\{0, \dots, r\}$ and designates its $(u+1)$st element as a. Choosing the smaller and larger elements gives the summand. Summing $\binom{j}{u}$ yields the first term of (3.1).

For a short prefix of length $\ell < j$, the retained list is the whole current chain, so its last checkpoint is exactly r. For $r > 0$, choose $\ell - 1$ earlier checkpoints and one of two port bindings at each event. This gives $2^\ell \binom{r-1}{\ell-1}$. Its side increments determine one carrier arc; summing across arcs therefore counts it once. The empty chain occurs only at $r = 0$.

Every counted prefix is actual at every placement compatible with its fixed carrier arc. For a saturated prefix, append its remaining nonempty left and right increments in any fixed order, then apply the inherited complete-chain fibre theorem. For a short prefix, use its complete chain directly. Conversely an actual chain determines all the chosen endpoints and bindings. This proves both counting and attainability.

Lemma 3.2.
At a full carrier the number of retained labels is

$$T_{N,j} = 2^j \binom{N}{j} + \sum_{\ell=1}^{j-1} 2^\ell \binom{N-1}{\ell-1}. \tag{3.4}$$

Proof. *A saturated prefix chooses j checkpoints from $1, \dots, N$ and their j bindings. It can be completed to the full carrier if it does not already end there. A short chain must end at N, giving the second term. The actual chain theorem realizes these choices. With two packets left, absorbing the whole complement in port direction L or R produces different ordered records, so both choices remain in the count. Physical terminal phase is not part of the retained label.*

Theorem 3.3.
Define

$$A_n^{[j]} = n \sum_{r=0}^{N-1} c_{r,j} \binom{2N-r}{N-r}. \tag{3.5}$$

For every $n \geq 3$ and $j \geq 1$,

$$Q_n^{[j]} = 1 + R_n + T_{N,j} + A_n^{[j]}, \qquad Q_{n,\mathrm{reg}}^{[j]} = T_{N,j} + A_n^{[j]}. \tag{3.6}$$

The normal-form keys are: one key for permanently unregistrable states; P for registerable unregistered states; (P, L_j) for registered nonterminal states; and L_j for registered terminals.

More explicitly, let $\mathcal{R}_n$ be the actual registerable unregistered placements, $\mathcal{T}_{N,j}$ the actual terminal label image, and $\mathcal{P}(C, L)$ the actual placements compatible with a proper carrier C and retained label L. There is an equivalence

$$X_n/{\equiv_j} \;\simeq\; \{\star\} \;\sqcup\; \mathcal{R}_n \;\sqcup\; \mathcal{T}_{N,j} \;\sqcup \coprod_{C \text{ proper}} \; \coprod_{L \text{ compatible with } C} \mathcal{P}(C, L).$$

The parameter spaces use actual attainable labels and literal placements. Equality in this normal form is equivalent to the all-word observer equivalence, and the two encoding/decoding composites are identities. Decoding chooses a realizing history but returns its observer class; no uniqueness of the raw word is required.

Proof. *Apply (2.4) to the inherited future classes. In registered nonterminal classes, different placements remain separated by a future suffix, and different labels are separated by the current report. Equal placement and label give both equal future behaviour and equal current reports. At rank one there are no future selected events, so equality is exactly equality of labels. Registered labels differ from $\perp$, separating these terminal classes from the dead unregistered class. The inherited preregistration classification is unchanged.*

For a fixed proper carrier and compatible label, Lemma 3.1 supplies exactly the entire placement fibre of size $n\binom{2N-r}{N-r}$. This counts placements over a label, not histories over one placement. Summing labels over carrier arcs gives (3.5). Lemma 3.2 supplies the terminal count, and the other two regions give R_n and one. Restricting to the registered domain removes just those two regions.

For $r \geq 1$, the first two depths give

$$c_{r,1} = r(r+1), \qquad c_{r,2} = 2 + \frac{2}{3} r(r^2 - 1), \tag{3.7}$$

whereas $c_{0,j} = 1$ at every depth. The terminal counts are $T_{N,1} = 2N$ and $T_{N,2} = 2 + 2N(N-1)$.

Corollary 3.4.

Let s_n be the complete registered-summary count. For $j \geq N$,

$$Q_n^{[j]} = 1 + R_n + s_n - (n-1)\,2\,3^{N-1}. \tag{3.8}$$

Proof. *Every chain is now retained in full. There are $2\,3^{N-1}$ complete chains ending at checkpoint N. Each has n phases in a complete summary and one terminal prefix key here. All nonterminal placements remain literal. The only subtraction is therefore $n-1$ per full-carrier chain.*

Corollary 3.5.

For $1 \leq j < k$, the identity on actual states induces a surjection $X_n/{\equiv_k} \twoheadrightarrow X_n/{\equiv_j}$. Both the whole-domain and registered-domain counts strictly increase from depth j to $j+1$ whenever $1 \leq j < N$. Their equivalence relations stabilize for $j \geq N$.

Proof. *The depth-j label is recovered from the depth-k label by truncating the records and recomputing the short/saturated marker. The future-event contract is unchanged, so (2.4) gives $x \equiv_k y \Rightarrow x \equiv_j y$.*

For strictness, take the same first j singleton R absorptions, then absorb the entire remaining complement with final port binding L or R. Since $j < N$, the carrier before that last event is proper. The inherited chain realization theorem supplies both actual terminal states. Their depth-j labels agree and their futures are silent, but their depth-$(j+1)$ labels contain different complete final records. They therefore give two distinct finer classes inside one coarser class, in either state domain.

Every chain has length at most N. For $j \geq N$ its label determines the whole chain, and the marker is determined by that chain and j. Equality of labels consequently has the same meaning at every such depth. Equation (2.4) proves stabilization.

4 Fixed-Depth Asymptotics

Lemma 4.1.

For each fixed $j \geq 1$,

$$\sum_{r \geq 0} c_{r,j} 2^{-r} = 5\,2^j - 1. \tag{4.1}$$

Proof. *For $|x| < 1$ the two relevant sums are*

$$\sum_{r \geq 1} \binom{r+1}{j+1} x^r = \frac{x^j}{(1-x)^{j+2}}, \qquad \sum_{r \geq \ell} \binom{r-1}{\ell-1} x^r = \left(\frac{x}{1-x}\right)^{\ell}. \tag{4.2}$$

They follow by differentiating the geometric series, or by the ordinary generating function for compositions. At $x = 1/2$, the empty-chain contribution is 1, the saturated contribution is 2^{j+2}, and the short contributions sum to $2^j - 2$.

Theorem 4.2.

For fixed $j \geq 1$, as $n \to \infty$,

$$Q_n^{[j]} \sim \frac{5\,2^j}{16\sqrt{\pi}} \sqrt{n}\, 4^n, \qquad Q_{n,\mathrm{reg}}^{[j]} \sim \frac{5\,2^j - 1}{16\sqrt{\pi}} \sqrt{n}\, 4^n. \tag{4.3}$$

Proof. *Put $B_N = \binom{2N}{N}$. For $0 \leq r \leq N$,*

$$\frac{\binom{2N-r}{N-r}}{B_N} = \prod_{i=0}^{r-1} \frac{N-i}{2N-i} \leq 2^{-r}, \tag{4.4}$$

and for fixed r this tends to 2^{-r}. The coefficients $c_{r,j}$ have polynomial growth in r for fixed j. Extending the summand by zero for $r \geq N$, (4.4) gives a summable bound, so dominated convergence and Lemma 4.1 imply

$$\frac{A_n^{[j]}}{nB_N} \longrightarrow 5\,2^j - 1. \tag{4.5}$$

The terminal contribution is polynomial in N at fixed depth and is negligible on this scale. Moreover $R_n/(nB_N) = N/(N+1) \to 1$. Stirling's formula [7] gives the central-binomial asymptotic $B_N \sim 4^N/\sqrt{\pi N}$, so

$$nB_N \sim \frac{\sqrt{n}\,4^n}{16\sqrt{\pi}}, \tag{4.6}$$

since $N = n-2$. Combining the contributions proves (4.3).

For example the registered coefficients are 9 and 19 at depths one and two, while the whole-domain coefficients are 10 and 20. The fixed-depth result does not supply a uniform estimate for growing j; that requires a separate argument.

5 A Uniform Truncated-Binomial Reduction

Set

$$S_{r,j} = \sum_{\ell=0}^{\min(j,r)} 2^\ell \binom{r}{\ell}, \qquad Z_{N,j} = \sum_{r=0}^{N-1} \binom{2N-r}{N-r} S_{r,j}. \tag{5.1}$$

Lemma 5.1.
For every $r \geq 0$ and $j \geq 1$,

$$\frac{S_{r,j}}{r+1} \leq c_{r,j} \leq (r+2)S_{r,j}. \tag{5.2}$$

Consequently

$$\frac{Z_{N,j}}{N} \leq \frac{A_n^{[j]}}{n} \leq (N+1)Z_{N,j}. \tag{5.3}$$

Proof. *At $r = 0$ both coefficients are one. For $r \geq 1$, the saturated term in (3.1) is at most $(r+1)2^j\binom{r}{j}$ when $j \leq r$, and is zero otherwise. Every short term is at most its corresponding term in $S_{r,j}$. This gives the upper bound.*

For $1 \leq \ell < j$ use $\binom{r-1}{\ell-1} = (\ell/r)\binom{r}{\ell}$. When $j \leq r$, also use $\binom{r+1}{j+1} \geq \binom{r}{j}$. Together these give $S_{r,j} - 1 \leq rc_{r,j}$, including $j > r$. There is at least one compatible label, so $c_{r,j} \geq 1$ and $S_{r,j} \leq (r+1)c_{r,j}$. Multiplying by the placement weight and using $r+1 \leq N$, $r+2 \leq N+1$ proves (5.3).

Lemma 5.2.
For all $N \geq 1$ and $j \geq 1$,

$$Z_{N,j} = \sum_{\ell=0}^{\min(j,N)} 2^\ell \left[\binom{2N+1}{N-\ell} - \binom{N}{\ell}\right]. \tag{5.4}$$

Proof. *Exchange the two finite sums in (5.1). For fixed ℓ, append the term $r = N$ and apply the convolution*

$$\sum_{r=\ell}^{N} \binom{r}{\ell}\binom{2N-r}{N} = \binom{2N+1}{N-\ell}. \tag{5.5}$$

To prove (5.5), choose an $(N+\ell+1)$-element subset of a $(2N+1)$-element ordered set, and let its $(\ell+1)$st member be at position $r+1$. Choose ℓ members before it and N after it. Complementing the chosen subset gives the right-hand binomial coefficient. The appended term is $\binom{N}{\ell}$.

Write $\mathcal{U}_{N,j} = \sum_{\ell\le\min(j,N)} 2^{\ell}\binom{2N+1}{N-\ell}$. The subtracted sum in (5.4) is at most 3^N, whereas $\mathcal{U}_{N,j}$ contains $\binom{2N+1}{N} \ge B_N \ge 4^N/(2N+1)$. Therefore

$$0 \le 1 - \frac{Z_{N,j}}{\mathcal{U}_{N,j}} \le (2N+1)\left(\frac{3}{4}\right)^N = o(1) \tag{5.6}$$

uniformly in positive j. Also $T_{N,j} \le 3^N$ and $R_n \le nB_N$. Since $Z_{N,j} \ge B_N$, equations (3.6), (5.3) and (5.6) show that $\mathcal{U}_{N,j}$ determines the exponential rate of both observer counts, with only polynomial factors and an exponentially negligible subtraction.

6 The Linear-Memory Growth Spectrum

Let $H(x) = -x\log x - (1-x)\log(1-x)$, with $0\log 0 = 0$. We use the elementary bounds

$$\frac{e^{MH(k/M)}}{M+1} \le \binom{M}{k} \le e^{MH(k/M)} \qquad (0 \le k \le M). \tag{6.1}$$

For $0 < k < M$, take success probability k/M in the binomial distribution. Its kth probability is a mode, at least $1/(M+1)$ and at most one. Writing that probability explicitly gives (6.1). The endpoint cases are immediate.

Theorem 6.1.

Fix $0 < \theta < 1$ and put $j_n = \lfloor \theta n \rfloor$ on the cofinite range where $j_n \ge 1$. Then both limits exist, and

$$\lim_{n\to\infty} (Q_n^{[j_n]})^{1/n} = \lim_{n\to\infty} (Q_{n,\mathrm{reg}}^{[j_n]})^{1/n} = \lambda(\theta), \tag{6.2}$$

where λ is given by (1.1). In particular $4 < \lambda(\theta) < 9/2$ for $0 < \theta < 1/3$.

Proof. *With $N = n-2$ and $\ell/N \to \beta$, the rate of the summand in $\mathcal{U}_{N,j}$ is*

$$G(\beta) = \beta\log 2 + 2H\left(\frac{1-\beta}{2}\right) = (2+\beta)\log 2 - (1+\beta)\log(1+\beta) - (1-\beta)\log(1-\beta). \tag{6.3}$$

Indeed (6.1) bounds its logarithm by $\ell\log 2 + (2N+1)H((N-\ell)/(2N+1))$, with error at most $\log(2N+2)$. After division by N this expression converges uniformly to $G(\ell/N)$: entropy is uniformly continuous on $[0,1]$, and the change in its argument is uniformly $O(1/N)$. There are at most $N+1$ summands. The logarithm of their sum differs from that of the largest by at most $\log(N+1)$.

Since $j_n/N \to \theta$, compactness gives the upper rate $\max_{0\le\beta\le\theta} G(\beta)$. Conversely every $0 \le \beta < \theta$ can be approached by an admissible integer ℓ_N; continuity supplies the endpoint by approximation. Thus upper and lower rates agree. More explicitly at the threshold $\theta = 1/3$, $\ell_N = \lfloor N/3 \rfloor \le \lfloor n/3 \rfloor$ is admissible and $\ell_N/N \to 1/3$.

Differentiation in the interior gives

$$G'(\beta) = \log\frac{2(1-\beta)}{1+\beta}. \tag{6.4}$$

Hence G increases strictly from $G(0) = \log 4$ up to $\beta = 1/3$, where $G(1/3) = \log(9/2)$, and decreases after that point. The constrained maximum is at $\beta = \min(\theta, 1/3)$. Exponentiation gives (1.1), and the uniform comparison of Section 5 transfers it to both quotients. Finally $N/n \to 1$, so the same limit uses the exponent $1/n$.

Corollary 6.2.

If $j_n \geq 1$ eventually and $j_n = o(n)$, both observer counts have exponential base 4.

Proof. *In the same uniform estimate every admissible $0 \leq \ell/N \leq j_n/N$ tends uniformly to zero. The maximum rate is therefore $G(0) = \log 4$, and the central summand gives the matching lower bound.*

The full rate is attained at memory density $1/3$, even though many longer histories remain unidentified by that prefix. The next result separately establishes asymptotic equality of counts above that density.

Lemma 6.3.

Let $h_{n,\ell}$ count complete registered summaries whose selected chain has length ℓ. For $0 \leq \ell \leq N$,

$$h_{n,\ell} = n\, 2^{\ell} \binom{2N}{N-\ell}, \qquad s_n = n \sum_{\ell=0}^{N} 2^{\ell} \binom{2N}{N-\ell}. \tag{6.5}$$

Proof. *At $\ell = 0$ the carrier is F, and its placement fibre has size $n\binom{2N}{N}$. For $\ell \geq 1$, group complete chains by their last checkpoint r. There are $2^{\ell}\binom{r-1}{\ell-1}$ chains, each with $n\binom{2N-r}{N-r}$ placements. This includes $r = N$, where the fibre is the n full-carrier phases. The required convolution is*

$$\sum_{r=\ell}^{N} \binom{r-1}{\ell-1} \binom{2N-r}{N-r} = \binom{2N}{N-\ell}. \tag{6.6}$$

Choose an $(N+\ell)$-element subset of a $2N$-element ordered set and designate its ℓth member as position r. Choose $\ell - 1$ members before it and N after it. Complementation gives the right-hand side. The inherited actual chain/fibre classification therefore proves (6.5) for actual summaries.

There is a useful exact reweighting. If K has the auxiliary binomial distribution $\mathrm{Bin}(2N, 1/3)$, then

$$\frac{h_{n,\ell}}{n(9/2)^N} = \Pr(K = N - \ell), \qquad \frac{s_n}{n(9/2)^N} = \Pr(K \leq N). \tag{6.7}$$

These identities describe a finite counting sum, not a random choice of execution actions. Since

$$\Pr(K > N) \leq 2^{-N}\mathbb{E}[2^K] = 2^{-N}(4/3)^{2N} = (8/9)^N, \tag{6.8}$$

we obtain the explicit lower bound $s_n \geq n(9/2)^N[1 - (8/9)^N]$.

Corollary 6.4.

Fix $1/3 < \theta < 1$ and $j_n = \lfloor \theta n \rfloor$. Then

$$\frac{Q_n^{[j_n]}}{Q_n^{\mathrm{full}}} \longrightarrow 1, \qquad \frac{Q_{n,\mathrm{reg}}^{[j_n]}}{s_n} \longrightarrow 1. \tag{6.9}$$

More explicitly, fix any ρ with $1/3 < \rho < \theta$ and put $\delta_\rho = \log(9/2) - G(\rho) > 0$. For all sufficiently large n, each relative deficit in (6.9) is nonnegative and at most

$$\frac{(N+1)e^{-N\delta_\rho} + \frac{2}{3}(2/3)^N}{1-(8/9)^N}. \tag{6.10}$$

Proof. *Let $Q_n^{[\infty]}$ denote the full-prefix quotient of Corollary 3.4. The map from full-summary classes to full-prefix classes forgets only terminal phase. Its exact loss is*

$$D_n = (n-1)\,2\,3^{N-1}. \tag{6.11}$$

Under the further map to depth-j classes, every full-prefix class with chain length at most j has a singleton fibre. Below depth j the short label recovers the whole chain. At length exactly j, the last retained record already gives the current carrier. A longer chain with that prefix would strictly enlarge the carrier, so it cannot have the same nonterminal placement; at a full carrier no further absorption is possible. Thus these classes do not merge with each other or with any longer-history class.

Set $L_{n,>j} = \sum_{\ell=j+1}^{N} h_{n,\ell}$, with an empty sum equal to zero. The number of remaining full-prefix classes is at most $L_{n,>j}$. Consequently

$$0 \le Q_n^{\mathrm{full}} - Q_n^{[j]} \le D_n + L_{n,>j}. \tag{6.12}$$

The same loss equals $s_n - Q_{n,\mathrm{reg}}^{[j]}$, because both whole-domain counts contain the identical $1 + R_n$ contribution.

Eventually $j_n \ge \rho N$. By (6.1), for $\beta = \ell/N$, $2^\ell \binom{2N}{N-\ell} \le e^{NG(\beta)}$. The function G decreases strictly on $(1/3, 1]$, so

$$L_{n,>j_n} \le n(N+1)e^{NG(\rho)}. \tag{6.13}$$

Divide (6.12) by $Q_n^{\mathrm{full}} \ge s_n$, or by s_n for the registered ratio. Use (6.8) and $D_n/[n(9/2)^N] \le \frac{2}{3}(2/3)^N$ to get (6.10). Both terms in its numerator decay exponentially, proving (6.9).

Distinct finite quotients can thus have asymptotically equal cardinality. For $0 < \theta < 1/3$, Theorem 6.1 and (6.7)–(6.8) instead give $Q_n^{[\lfloor \theta n \rfloor]}/Q_n^{\mathrm{full}} \to 0$, by their strictly different exponential bases. At the boundary, where the entropy gap in (6.13) vanishes, the exact prefix count gives a Gaussian transition.

Theorem 6.5.

Fix $u \in \mathbb{R}$ and put

$$j_n = \left\lfloor \frac{n}{3} + u\sqrt{n} \right\rfloor.$$

For all sufficiently large n, $1 \le j_n \le N$. Then

$$\boxed{\frac{Q_n^{[j_n]}}{Q_n^{\mathrm{full}}} \longrightarrow \Phi(3u/2), \qquad \frac{Q_{n,\mathrm{reg}}^{[j_n]}}{s_n} \longrightarrow \Phi(3u/2),} \tag{6.14}$$

where Φ is the standard normal distribution function. In particular,

$$\boxed{\frac{Q_n^{[\lfloor n/3 \rfloor]}}{Q_n^{\mathrm{full}}} \longrightarrow \frac{1}{2}.} \tag{6.15}$$

Moreover, for every fixed $A > 0$, with $j_n(u) = \lfloor n/3 + u\sqrt{n} \rfloor$, both limits hold uniformly at Berry–Esseen rate; for all sufficiently large n, the indices satisfy $1 \le j_n(u) \le N$ simultaneously for $|u| \le A$:

$$\boxed{\begin{aligned}\sup_{|u|\le A}\left|\frac{Q_n^{[j_n(u)]}}{Q_n^{\mathrm{full}}}-\Phi(3u/2)\right| &= O_A(n^{-1/2}),\\ \sup_{|u|\le A}\left|\frac{Q_{n,\mathrm{reg}}^{[j_n(u)]}}{s_n}-\Phi(3u/2)\right| &= O_A(n^{-1/2}).\end{aligned}} \tag{6.16}$$

Proof. *Fix $A>0$ and let $|u|\le A$ throughout. All estimates below are uniform over this interval. Write*

$$H_{<j}=\sum_{\ell=0}^{j-1}h_{n,\ell},\qquad U_{N,j}^{\mathrm{short}}=\sum_{\ell=1}^{j-1}2^{\ell}\binom{N-1}{\ell-1},$$

and

$$V_{N,j}=\binom{2N+1}{N-j}+\binom{2N+1}{N-j-1}-\binom{N}{j}-\binom{N}{j+1}.$$

Splitting the coefficient $c_{r,j}$ in (3.1) into its stopped and saturated-prefix terms, and applying (6.6), (5.5), and Pascal's identity, gives the exact identity for $1\le j\le N$:

$$Q_{n,\mathrm{reg}}^{[j]}=H_{<j}-(n-1)U_{N,j}^{\mathrm{short}}+n2^jV_{N,j}+2^j\binom{N}{j}. \tag{6.17}$$

For the stopped terms, the convolution is (6.6); the terminal stopped labels restore one copy of $U_{N,j}^{\mathrm{short}}$. For the saturated terms, use $\binom{r+1}{j+1}=\binom{r}{j+1}+\binom{r}{j}$ in (5.5), append the $r=N$ terms, and subtract those two endpoint terms. This yields exactly $V_{N,j}$.

In the window $j=N/3+O_A(\sqrt{N})$, the binomial local estimate gives, uniformly,

$$2^j\binom{2N+1}{N-j}+2^j\binom{2N+1}{N-j-1}=O_A\left(\frac{(9/2)^N}{\sqrt{N}}\right).$$

Indeed, with $x=(N-j)/(2N+1)$, Taylor expansion at $x=1/3$ gives $j\log 2+(2N+1)H(x)=N\log(9/2)+O_A(1)$: the order-$\sqrt{N}$ terms cancel because $H'(1/3)=\log 2$. Stirling's local prefactor then contributes $O(N^{-1/2})$. The adjacent binomial coefficient has the same estimate, since its argument shifts by only $O(N^{-1})$. Also $U_{N,j}^{\mathrm{short}}\le 2\,3^{N-1}$ and both endpoint terms $2^j\binom{N}{j}$ and $2^j\binom{N}{j+1}$ are at most 3^N. After multiplication by their coefficients in (6.17), these terms divided by s_n are $O((2/3)^N)$. Since $s_n\sim n(9/2)^N$, (6.17) implies

$$\frac{Q_{n,\mathrm{reg}}^{[j]}}{s_n}=\frac{H_{<j}}{s_n}+O_A(N^{-1/2}).$$

The denominator replacement is valid at this rate: (6.7)–(6.8) give $s_n/[n(9/2)^N]=1+O((8/9)^N)$. The whole-domain ratio differs from the registered-domain ratio by $O((R_n+1)/s_n)$, uniformly in u, because $0\le Q_{n,\mathrm{reg}}^{[j]}\le s_n$. The exact formula $R_n=n\binom{2N}{N-1}$ and Stirling's estimate give $R_n/s_n=O(N^{-1/2}(8/9)^N)$.

Let $K\sim\mathrm{Bin}(2N,1/3)$. By (6.7), $H_{<j}/s_n=\Pr(K>N-j\mid K\le N)$. The complement of the conditioning event has probability at most $(8/9)^N$ by (6.8), so conditioning changes this tail by an exponentially small amount. The Berry–Esseen inequality [1, 4] for the standardized binomial variable gives a uniform $O(N^{-1/2})$ error. Its threshold satisfies

$$\frac{N-j_n(u)-2N/3}{(2/3)\sqrt{N}}=-\frac{3u}{2}+O_A(N^{-1/2}),$$

where the error includes the floor in $j_n(u)$ and $\sqrt{n}-\sqrt{N}=O(N^{-1/2})$. Since Φ is Lipschitz, this proves $H_{<j_n(u)}/s_n=\Phi(3u/2)+O_A(N^{-1/2})$. Combining with the prefix correction proves (6.14) and (6.16); setting $u=0$ gives (6.15).

7 Formal Verification and Local Controls

7.1 Finite Structure and Counting

The separate Lean 4 companion in `experiments/paper44/lean` formalizes the concrete singleton-seed observer and its actual normal forms, including all-parameter reachability and inverse identities. Its principal manuscript interfaces are:

Manuscript result	Lean interface
Equation (2.4) and region classification	`circular_prefix_classification`
Lemma 3.1, including $r = 0$	`card_actualProperLabelsAt_all`
Lemma 3.2 on actual terminal labels	`terminalPrefixActualEquiv`, `card_actualTerminalLabel_paper`
Fixed-carrier/label placement fibres	`properPrefixPlacementEquiv`, `card_actualProperPlacements_paper`
Theorem 3.3, actual full and registered normal forms	`prefixObserverNormalEquiv`, `registeredPrefixObserverNormalEquiv`
Theorem 3.3, both cardinality formulas	`paperTheorem33`

These results hold for every $n \geq 3$ and $j \geq 1$ and use the complete ordered records and null-preserving per-letter traces of Section 2. The companion reuses Paper XL's abstract observer API and the fixed-pair and future-event companions' execution and classification theorems. The new finite work concerns actual prefix-label images, their compatible placement fibres, and their assembly into the minimal quotient.

The normal-form equivalences are noncomputable and use classical choice. They neither require a unique realizing raw word nor extract an executable minimizer. The axiom audit records only the standard dependencies `propext`, `Classical.choice` and `Quot.sound`, with no custom axioms or proof placeholders.

The project pins Lean 4.33.0 and Mathlib revision `db584cd6` (the full pin is in the companion's Lake manifest). With the declared sibling source dependencies present, its validation commands are:

```
lake --no-cache build
lake env lean Checks.lean
lake env lean AxiomAudit.lean
```

The local structure-and-counting milestone records exact proof-source hashes and dependency identity separately from this evolving manuscript. Its scope stops at the finite classification and Theorem 3.3. Corollaries 3.4–3.5 and the analytic results in Sections 4–6 remain paper proofs; the growth spectrum and Gaussian window are not claimed as Lean-certified. Reusing a local dependency cache is not independent cold-start reproduction.

7.2 Bounded Computational Controls

The local control first generates the complete finite actual state graph, then independently of the counting formulas minimizes it for each declared prefix depth. It compares the entire equivalence partition with the keys in Theorem 3.3, not just the total number. For $3 \leq n \leq 7$ and $1 \leq j \leq n-1$ all 20 cases agree.

n	Reachable states	Minimal counts at depths $1, \dots, n-1$
3	30	12, 12
4	176	69, 71, 71
5	920	402, 410, 414, 414
6	4,566	2,085, 2,283, 2,303, 2,311, 2,311
7	22,134	9,965, 11,803, 12,363, 12,411, 12,427, 12,427

The saved controls also check recovery of the earlier depth-one, depth-two and full-prefix formulas, and finite evaluations of the rate formula. These calculations share the local consumer transition code and serve as bounded consistency controls. The all-parameter count and limits are proved in Sections 3–6 using the two imported structural classifications.

Separate exact-arithmetic tests check (6.5)–(6.8), the finite deficit inequality (6.12), and the decomposition (6.17). They do not extend the earlier saved generated-domain replay or supply the asymptotic argument.

8 Conclusion

Retained prefix memory gives an exact family of minimal observers, with actual history labels and placement fibres. Fixed and sublinear depth preserve exponential base 4, whereas linear depth produces strictly intermediate rates before the base reaches $9/2$. At the threshold, the state-count ratio follows the Gaussian profile $\Phi(3u/2)$ across a window of width $\sqrt{n}$; at its center the ratio tends to $1/2$. The threshold concerns the growth of the minimal state space, rather than recovery of every historical record. Strictly above the threshold, the prefix quotient has asymptotically the full-summary state count, despite continuing to identify some distinct complete summaries.

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